

Problem 1 : Detection of Backdoor Attacks using HPCs

Objective: In backdoor attack an untrusted party can train a model in such a way that it behaves normally on most of the inputs, except when there is a special pattern embedded in the input, which can be exploited to selectively change the model predictions. The teams will be provided with hardware performance counters (HPC) measured from benign input inferences, and they have to train an anomaly detector model on those traces, so that when the detector sees traces from a backdoor inference it can label it differently from the clean inputs.

Trace File Format: (link)

- A .csv file containing HPC traces collected from benign inferences.
- The column names indicate the name of the particular HPC.
- There are 800 rows each for one clean inference.
- No trace for backdoor inference is provided to maintain a strict anomaly detection scenario.

Deliverables: A GitHub repository containing:

- The complete code, with finalized parameters.
- A saved detector model.
- A README file for documentation and a short report (.pdf) describing the method.

Evaluation:

- The submissions will be evaluated on a private validation set containing traces from clean and backdoor runs.
- Standard metrics such as Accuracy, TPR, FPR, F1, AUROC will be used.

Problem 2 : *Black Box Adversarial Attack on CNN*

Objective: The teams will be provided with a pretrained Convolution Neural Network (CNN) and collection of input images and they have to perform black box attack on the CNN to generate one adversarial image against each of the input images. An adversarial image is almost identical to a “clean” image and differs only by a small amount of visually imperceptible noise, but causes the model to give incorrect prediction. A black box attack means the attacker is allowed to query the CNN at arbitrary input images and observe the output labels, but cannot look at the CNN architecture, parameters, and intermediate outputs including prediction scores.

Shared Files: ([link](#))

- A .zip file containing 1000 clean images (.png) and a pretrained CNN model (.pt).
- A partially complete notebook (.ipynb) containing standard code snippets for accessing the model and additional instructions.

Deliverables: A GitHub repository containing:

- The complete attack code in the shared notebook with finalized attack parameters.
- One adversarial image (.png) corresponding to each clean image. (1000 adversarial images in total).
- A README file for documentation and a short report (.pdf) describing the method.

Evaluation:

- Attack Success Rate (50%): The percentage of adversarial images that successfully change the model predictions.
- Imperceptibility (50%): The overall invisibility of the noise.
- Tie Breaking: Among the submission performing equally well in the above criteria, the one using a smaller number of model inferences and causing higher change in the prediction confidence will be ranked higher.